

What Factors are Most Associated with Heart Disease?

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Claim

Although age naturally increases the risk of heart disease, lifestyle choices such as smoking, heavy alcohol use, high cholesterol, and high blood pressure are major preventable factors that can significantly increase the likelihood of developing heart disease.

Importance

This topic is important because heart disease is the number one killer in the U.S killing over 680,000 people a year.

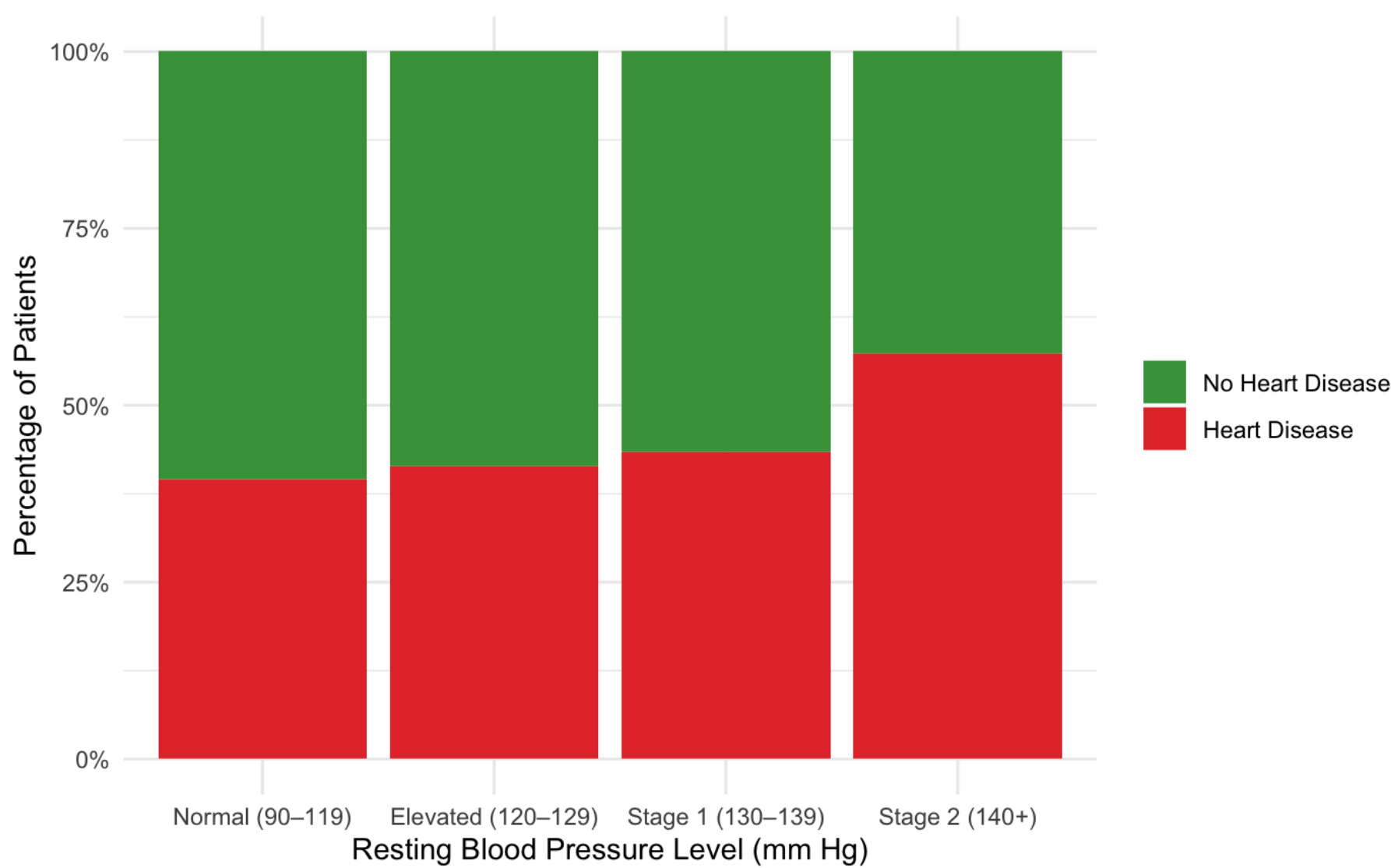
The data set suggests that age and cholesterol levels are important factors in heart disease risk. As individuals age, the impact of high cholesterol becomes more significant, increasing the like likelihood of cardiovascular disease

Data and Methods

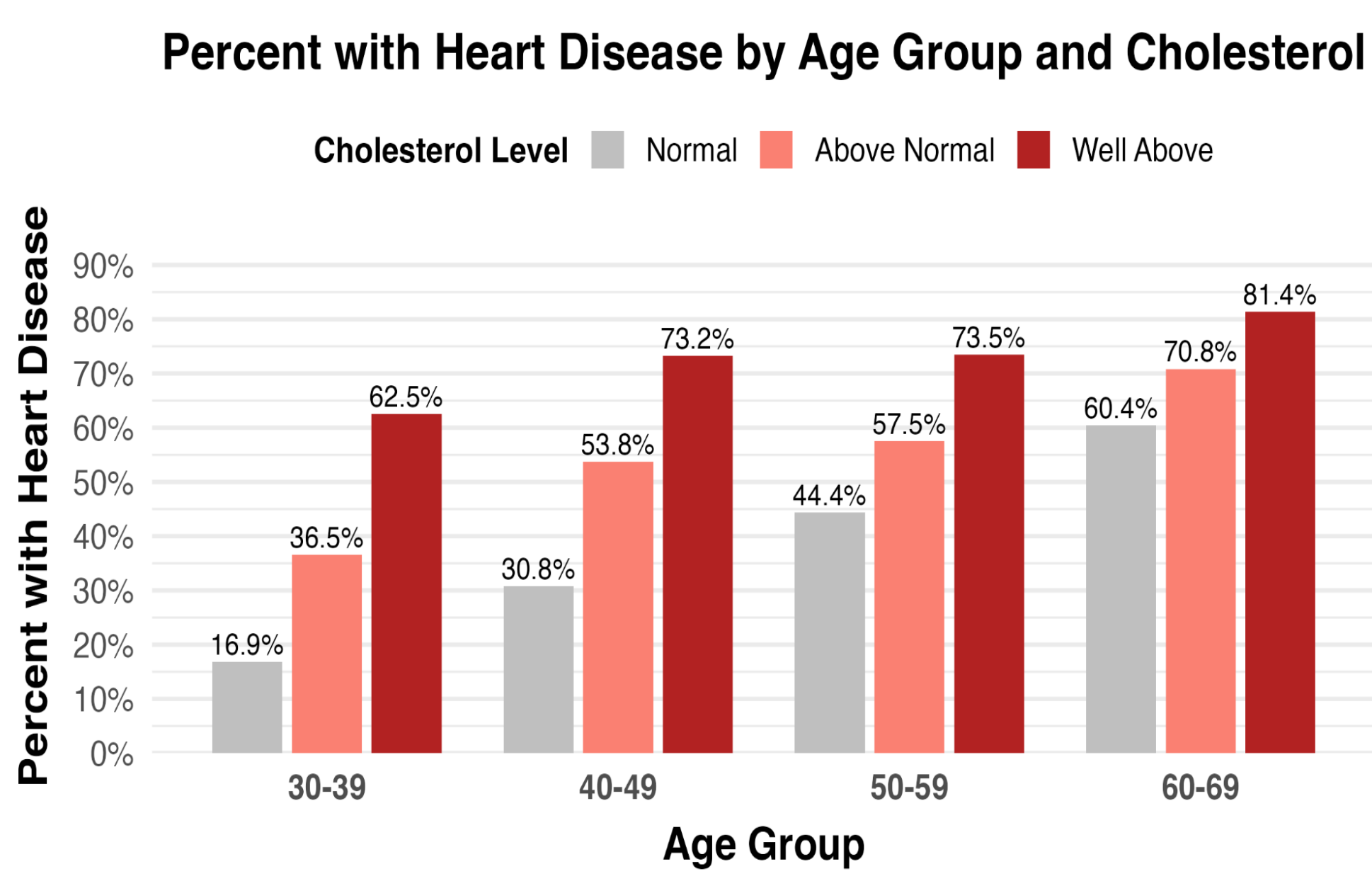
Our first dataset was created by Andras Janosi, William Steinrbunn, Matthias Pfisterer, and Robert Detano in 1989. The dataset was derived from 303 test results from patients undergoing angiography at the Cleveland Clinic. There are various variables within the dataset such as sex, age, and cholesterol(chol), however some other less straightforward ones are resting blood pressure(trestbps). The num variable is a categorical variable that determines whether the patient has heart disease or not. The second dataset was created in 2020 by data scientist, Svetlana Ulianova, it consists of 70000 test results of patients with additional various lifestyle profiles such as smoking and drinking.

Graphs

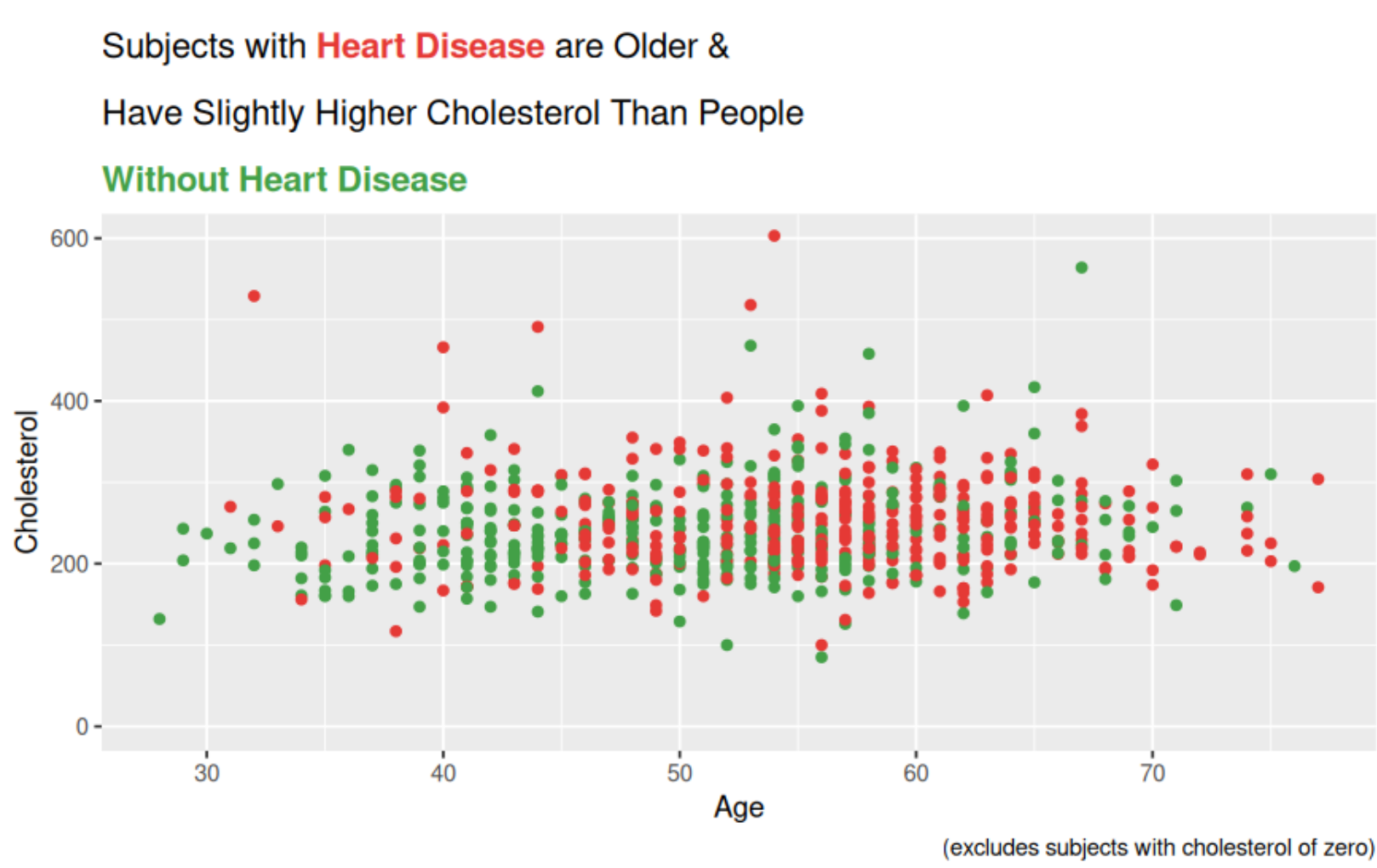
DATA SET 1989



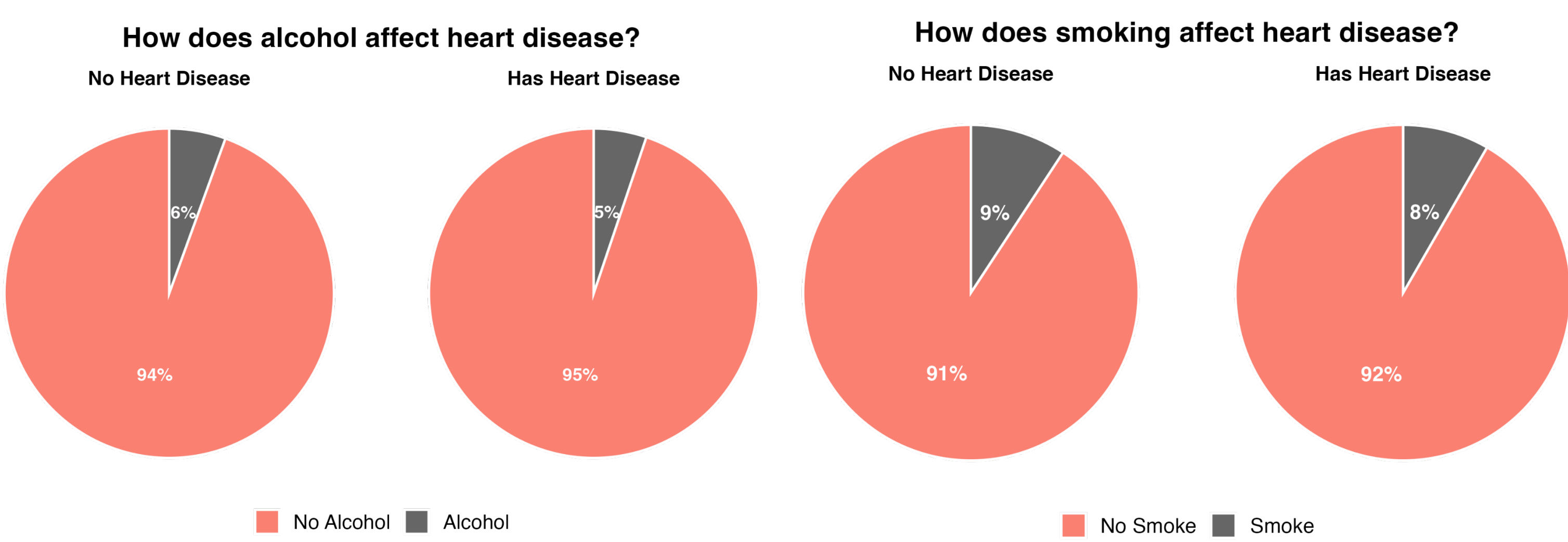
DATA SET 2020



DATA SET 1989



DATA SET 2020



CONCLUSIONS

Heart Disease is strongly associated with older age, high cholesterol, high blood pressure. Smoking did not appear to be a significant risk factor from our data. People with these risk factors should have heart disease in mind as a potential health threat.

SOURCES

Dataset 1989: Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989) Heart Disease [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C52P4X>

Dataset 2020: Aiaiai, David. (2020). Cardiovascular Diseases Dataset. <https://www.kaggle.com/datasets/aiaiaidavid/cardio-data-dv13032020/data>